

UNIVERSIDAD TECNOLÓGICA DE PANAMÁ

Facultad de Sistemas Computacionales

Asignatura: Programación II

Examen Final-Proyecto

Profesor: Napoleón Ibarra

Estudiante: Magdalena González **Cédula:** 4-819-1590

Estudiante: Adrián Jiménez **Cédula:** 4-839-2413

I PARTE. APPS CON CONEXIÓN-BASE DE DATOS. Valor 35 puntos.

```
(venv) turismo@turismo-db:~/turismo_chiriqui$ sudo systemctl status turismo
[sudo] password for turismo:
● turismo.service - Turismo Chiriqui Flask App (gunicorn)
   Loaded: loaded (/etc/systemd/system/turismo.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-07-12 15:26:29 EST; 1h 16min ago
     Main PID: 46558 (gunicorn)
       Tasks: 6 (limit: 5210)
      Memory: 109.2M
         CPU: 15.739s
    CGroup: /system.slice/turismo.service
            └─46558 /home/turismo/turismo_chiriqui/venv/bin/python3 /home/turismo/turismo_chiriqui/venv/bin/python3 /home/turismo/turismo_chiriqui/venv/bin/python3 /home/turismo/turismo_chiriqui/venv/bin/python3 /home/turismo/turismo_chiriqui/venv/bin/python3 /home/turismo/turismo_chiriqui/venv/bin/python3

Jul 12 15:26:29 turismo-db systemd[1]: Started turismo.service - Turismo Chiriqui Flask App (gunicorn).
Jul 12 15:26:29 turismo-db gunicorn[46558]: [2026-07-12 15:26:29 -0500] [46558] [INFO] Starting gunicorn 20.1.0
Jul 12 15:26:29 turismo-db gunicorn[46558]: [2026-07-12 15:26:29 -0500] [46558] [INFO] Listening at: /home/turismo/turismo_chiriqui/venv/bin/python3 /home/turismo/turismo_chiriqui/venv/bin/python3 /home/turismo/turismo_chiriqui/venv/bin/python3 /home/turismo/turismo_chiriqui/venv/bin/python3 /home/turismo/turismo_chiriqui/venv/bin/python3 /home/turismo/turismo_chiriqui/venv/bin/python3
Jul 12 15:26:29 turismo-db gunicorn[46569]: [2026-07-12 15:26:29 -0500] [46569] [INFO] Booting worker with pid: 46569
Jul 12 15:26:29 turismo-db gunicorn[46570]: [2026-07-12 15:26:29 -0500] [46570] [INFO] Booting worker with pid: 46570
Jul 12 15:26:29 turismo-db gunicorn[46571]: [2026-07-12 15:26:29 -0500] [46571] [INFO] Booting worker with pid: 46571
Jul 12 15:26:30 turismo-db gunicorn[46572]: [2026-07-12 15:26:30 -0500] [46572] [INFO] Booting worker with pid: 46572
Jul 12 15:26:30 turismo-db gunicorn[46558]: [2026-07-12 15:26:30 -0500] [46558] [INFO] Worker exiting (pid: 46558)

(venv) turismo@turismo-db:~/turismo_chiriqui$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-07-12 15:35:05 EST; 1h 7min ago
     Docs: man:nginx(8)
   Process: 49628 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Process: 49629 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Process: 50234 ExecReload=/usr/sbin/nginx -g daemon on; master_process on; -s reload (code=exited, status=0/SUCCESS)
     Main PID: 49662 (nginx)
       Tasks: 5 (limit: 5210)
      Memory: 4.1M
         CPU: 625ms
    CGroup: /system.slice/nginx.service
            └─49662 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
               50235 "nginx: worker process"

Loading remote monitoring, please wait...
```

```

192.168.56.107
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help X server Exit

(venv) turismo@turismo-db:~/turismo_chiriqui$ docker inspect mongo --format 'RestartPolicy: {{.HostConfig.RestartPolicy.Name}}'
RestartPolicy: unless-stopped
(venv) turismo@turismo-db:~/turismo_chiriqui$ docker ps --filter name=mongo
CONTAINER ID   IMAGE      COMMAND                  NAMES                  CREATED        STATUS        PORTS
ad0c1afa8540   mongo:4.4   "docker-entrypoint.s..."  mongo                  14 hours ago   Up 3 hours    0.0.0.0:27017->27017/tcp
(venv) turismo@turismo-db:~/turismo_chiriqui$ sudo iptables -L INPUT -n --line-numbers | head -10
Chain INPUT (policy ACCEPT)
num  target      prot opt source                destination            tcp dpts:40000:50000
1    ACCEPT      6    --  0.0.0.0/0             0.0.0.0/0              tcp dpt:21
2    ACCEPT      6    --  0.0.0.0/0             0.0.0.0/0              tcp dpt:22
3    ACCEPT      6    --  192.168.168.155       0.0.0.0/0              tcp dpt:22
4    ACCEPT      0    --  0.0.0.0/0             0.0.0.0/0
5    ACCEPT      6    --  192.168.56.0/24       0.0.0.0/0              tcp dpt:22
6    ACCEPT      6    --  192.168.56.0/24       0.0.0.0/0              tcp dpt:21
7    DROP        6    --  0.0.0.0/0             0.0.0.0/0              tcp dpt:22
8    DROP        6    --  0.0.0.0/0             0.0.0.0/0              tcp dpt:21
(venv) turismo@turismo-db:~/turismo_chiriqui$ sudo systemctl status knockd
● knockd.service - Port-Knock Daemon
   Loaded: loaded (/etc/systemd/system/knockd.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-07-12 14:36:26 EST; 2h 7min ago
     Docs: man:knockd(1)
    Main PID: 28509 (knockd)
      Tasks: 1 (limit: 5210)
     Memory: 464.0K
        CPU: 100ms
    CGroup: /system.slice/knockd.service
            └─28509 /usr/sbin/knockd -i enp0s8

Jul 12 14:36:26 turismo-db systemd[1]: Started knockd.service - Port-Knock Daemon.
Jul 12 14:36:26 turismo-db knockd[28509]: starting up, listening on enp0s8
Jul 12 14:36:46 turismo-db knockd[28509]: 192.168.168.155: openSSH: Stage 1
Jul 12 14:36:47 turismo-db knockd[28509]: 192.168.168.155: openSSH: Stage 2
Jul 12 14:36:47 turismo-db knockd[28509]: 192.168.168.155: openSSH: Stage 3
Jul 12 14:36:47 turismo-db knockd[28509]: 192.168.168.155: openSSH: OPEN SESAME
Jul 12 14:36:47 turismo-db knockd[28627]: openSSH: running command: /sbin/iptables -I INPUT
(venv) turismo@turismo-db:~/turismo_chiriqui$

Loading remote monitoring, please wait...
UNREGISTERED VERSION - Please support MobaXterm by subscribing to the professional edition here: https://mobaxterm.mobatek.net

```

192.168.56.112 (lili)

Terminal Sessions View X server Tools Games Settings Macros Help

Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help X server Exit

2. 192.168.56.112 (lili) 3. 192.168.56.107 4. 192.168.56.112 (lili)

```
turismo-db:~ # sudo systemctl status mariadb
● mariadb.service - MariaDB database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: disabled)
   Active: active (running) since Sun 2026-07-12 14:13:10 EDT; 3h 32min ago
     Docs: man:mysqld(8)
           https://mariadb.com/kb/en/library/systemd/
   Main PID: 1196 (mysqld)
    Status: "Taking your SQL requests now..."
     Tasks: 11 (limit: 3566)
      CPU: 27.307s
   CGroup: /system.slice/mariadb.service
           └─1196 /usr/sbin/mysqld --defaults-file=/etc/my.cnf --user=mysql --socket=/

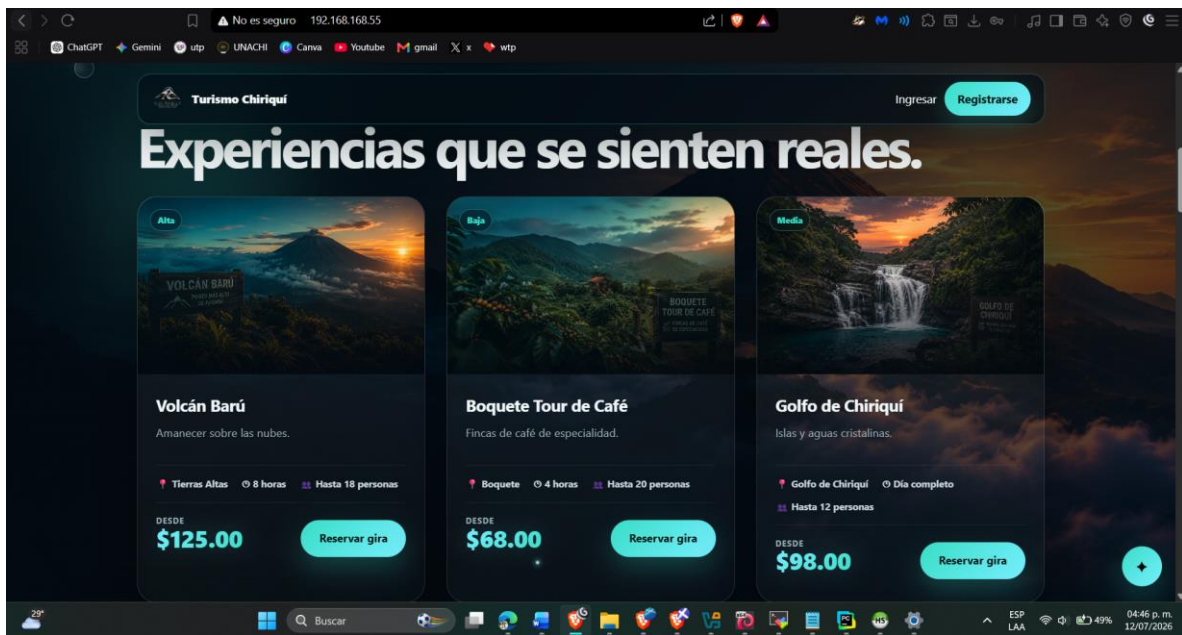
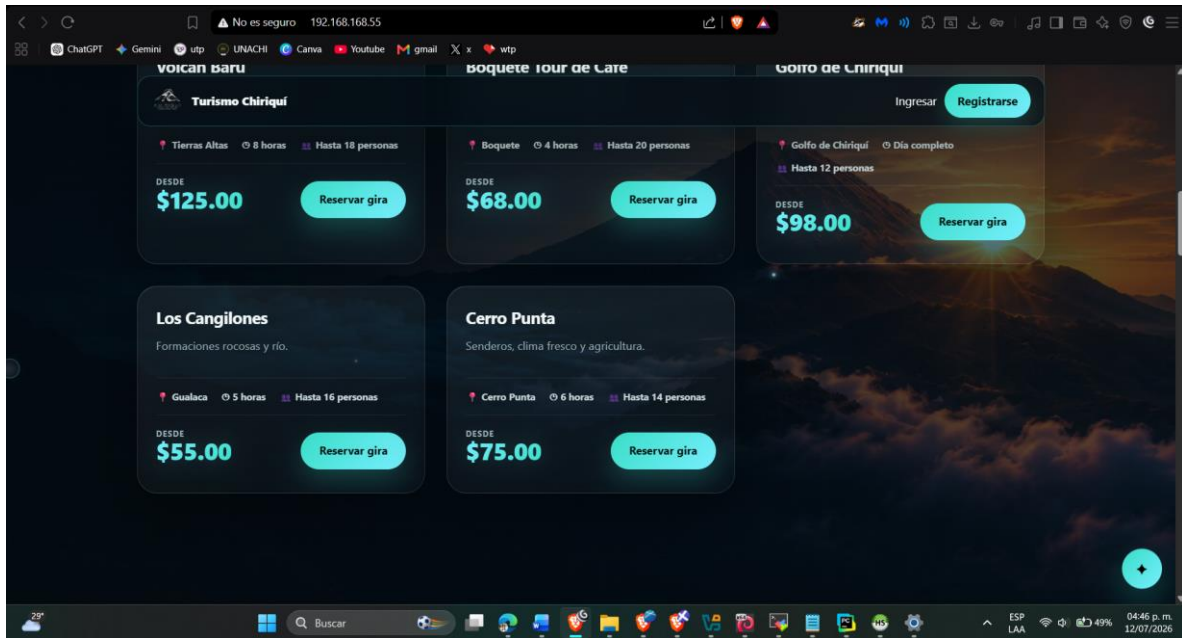
Jul 12 14:12:59 turismo-db systemd[1]: Starting MariaDB database server...
Jul 12 14:13:01 turismo-db mysql-systemd-helper[1196]: 2026-07-12 14:13:01 0 [Warning] f
Jul 12 14:13:10 turismo-db systemd[1]: Started MariaDB database server.
turismo-db:~ # sudo firewall-cmd --get-active-zones
sudo firewall-cmd --zone=dmz --list-all
sudo firewall-cmd --zone=public --list-all
dmz
  interfaces: eth3
docker
  interfaces: docker0
public (default)
  interfaces: eth1 eth2
turismo-db:~ # sudo systemctl status knockd
● knockd.service - Port-Knocking Daemon
   Loaded: loaded (/usr/lib/systemd/system/knockd.service; enabled; preset: disabled)
   Drop-In: /etc/systemd/system/knockd.service.d
           └─override.conf
   Active: active (running) since Sun 2026-07-12 17:24:37 EDT; 21min ago
     Main PID: 20125 (knockd)
       Tasks: 1 (limit: 3566)
        CPU: 3.097s
     CGroup: /system.slice/knockd.service
           └─20125 /usr/sbin/knockd -i eth3

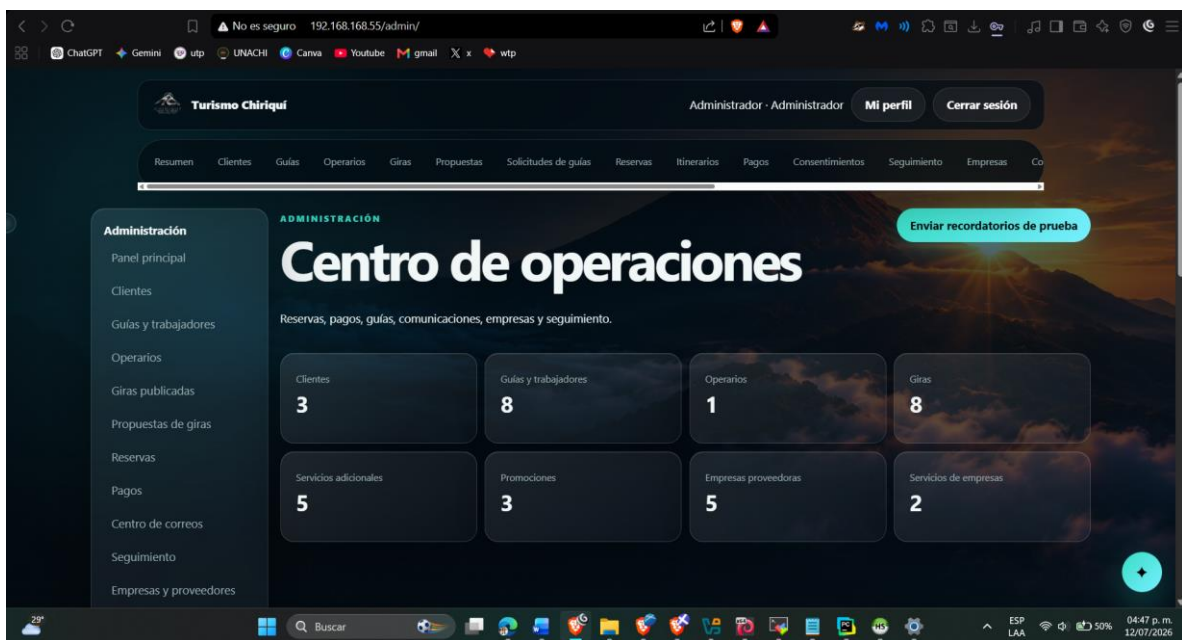
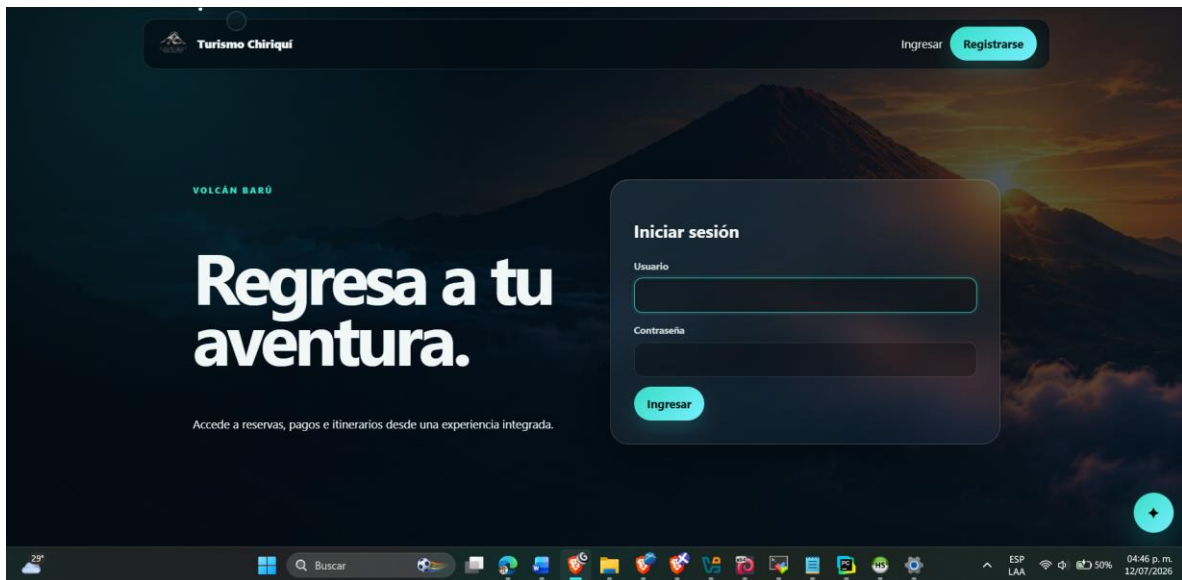
Jul 12 17:36:01 turismo-db knockd[25488]: success
Jul 12 17:36:52 turismo-db knockd[20125]: 192.168.168.155: opencloseSSH: Stage 1
Jul 12 17:36:53 turismo-db knockd[20125]: 192.168.168.155: opencloseSSH: Stage 2
```

turismo-db 1% 0.59 GB / 2.93 GB 0.01 Mb/s 0.00 Mb/s 213 min

UNREGISTERED VERSION - Please support MobaXterm by subscribing to the professional edition here: <https://mobaxterm.mobatek.net>

ESP LAA 49% 04:46 p. m. 12/07/2026





192.168.168.55/admin/crud/tours

Turismo Chiriquí

Administrador · Administrador

Mi perfil Cerrar sesión

Nombre	Destino	Descripción
Precio base	Duración	Cupos máximos
Mínimo de personas	Máximo de personas	Personas por guía
Precio de guía adicional	Dificultad	Proveedor
Promoción	Estado	Imagen
	activo	Seleccionar archivo Ningún archivo seleccionado

Crear registro Cancelar edición

Buscar en giras

Buscar

192.168.168.55/admin/crud/workers

Turismo Chiriquí

Administrador · Administrador

Mi perfil Cerrar sesión

Resumen Clientes Guías Operarios Giras Propuestas Solicitudes de guías Reservas Itinerarios Pagos Consentimientos Seguimiento Empresas Co

GESTIÓN ADMINISTRATIVA

Volver al resumen

Guías y trabajadores

Nombre completo	Correo	Teléfono
Cédula	Tipo	Especialidad
	activo	
Idiomas	Capacidad máxima recomendada	Experiencia
Disponibilidad	Estado de documentos	Observaciones Internas

Turismo Chiriquí

Administrador · Administrador

Mi perfil

Cerrar sesión

EMPRESA	SERVICIO	TIPO	PRECIO	ZONA	ESTADO
Transporte Barú S.A.	Transporte privado premium	transporte	\$55.00	David, Boquete y Volcán	Aprobado
Boquete Sabores	Almuerzo típico chiricano	alimentación	\$18.00	Boquete	Aprobado

Seguimiento por reserva

pendiente

#1 · Seguro de viaje por gira

Cliente: ADRIAN

Tour: Un camino a la prueba

Empresa: Aliado de seguros

Contacto: Por confirmar

Hora requerida

Estado

Por confirmar

pendiente

Servicio asignado por el cliente: pendiente

confirmado

#1 · Gira principal

Cliente: ADRIAN

Tour: Un camino a la prueba

Empresa: PRUEBAS

Contacto: magdalena.gonzalez@unachi.ac.pa

Hora requerida

Estado

14:19

pendiente

Reserva creada sobre una gira ofrecida por

pendiente

#2 · Transporte privado premium

Cliente: PRUEBAS

Tour: Volcán Barú

Empresa: Transporte Barú S.A.

Contacto: Por confirmar

Hora requerida

Estado

Por confirmar

pendiente

Servicio asignado por el cliente: pendiente

Turismo Chiriquí

Administrador · Administrador

Mi perfil

Cerrar sesión

Centro de Correos

Historial, errores SMTP, comprobantes y reenvíos.

Destinatario

Asunto

Mensaje

Enviar correo manual

Destinatario

Todos los tipos

Todos los estados

Filtrar

DESTINATARIO	ASUNTO	TIPO	ESTADO	FECHA	ERROR SMTP	ACCIONES
magdalena.gonzalez@unachi.ac.pa	Solicitud de empresa recibida	solicitud empresa	Enviado	2026-07-11 13:54:16		Reenviar Reenviar comprobante
liahwasat@gmail.com	Nueva empresa aliada pendiente	solicitud empresa	Enviado	2026-07-11 13:54:19		Reenviar Reenviar comprobante

192.168.168.55/operator/itineraries

Turismo Chiriquí Administrador · Administrador Mi perfil Cerrar sesión

Transporte turístico [https://maps.google.com/...](https://maps.google.com/)

Actividades del día

10:00 AM	Salida desde Dav	45 min	Punto de encue	Equipo Turismx	Proveedor si a	pendiente	Presentar docu	Quitar
10:45 AM	Llegada y prepar	30 min	Boquete / Volc	Guía asignado	Proveedor si a	pendiente	Revisión de eq	Quitar
11:15 AM	Experiencia princ	2 h	Ruta seleccio	Guía asignado	Proveedor si a	pendiente	Actividad ajust	Quitar
01:15 PM	Seguro de viaje g	45 min	Chiriquí	Aliado de segu	Aliado de segu	pendiente	Aplican condici	Quitar
02:00 PM	Retorno y cierre	60 min	David	Guía / Chofer	Proveedor si a	pendiente	Confirmación c	Quitar

Agregar actividad

Recomendaciones finales

Calzado cerrado, agua, protector solar, abrigo ligero y documento de identidad.

23°

192.168.168.55/admin/crud/complaints

Turismo Chiriquí Administrador · Administrador Mi perfil Cerrar sesión

Resumen Clientes Guías Operarios Giras Propuestas Solicitudes de guías Reservas Itinerarios Pagos Consentimientos Seguimiento Empresas Co

GESTIÓN ADMINISTRATIVA

Quejas

-- Volver al resumen

Asunto	Mensaje	Prioridad
Estado	Respuesta	
activo		

Crear registro Cancelar edición

23°

192.168.168.55/admin/crud/providers

Turismo Chiriquí

Administrador - Administrador [Mi perfil](#) [Cerrar sesión](#)

[-- Volver al resumen](#)

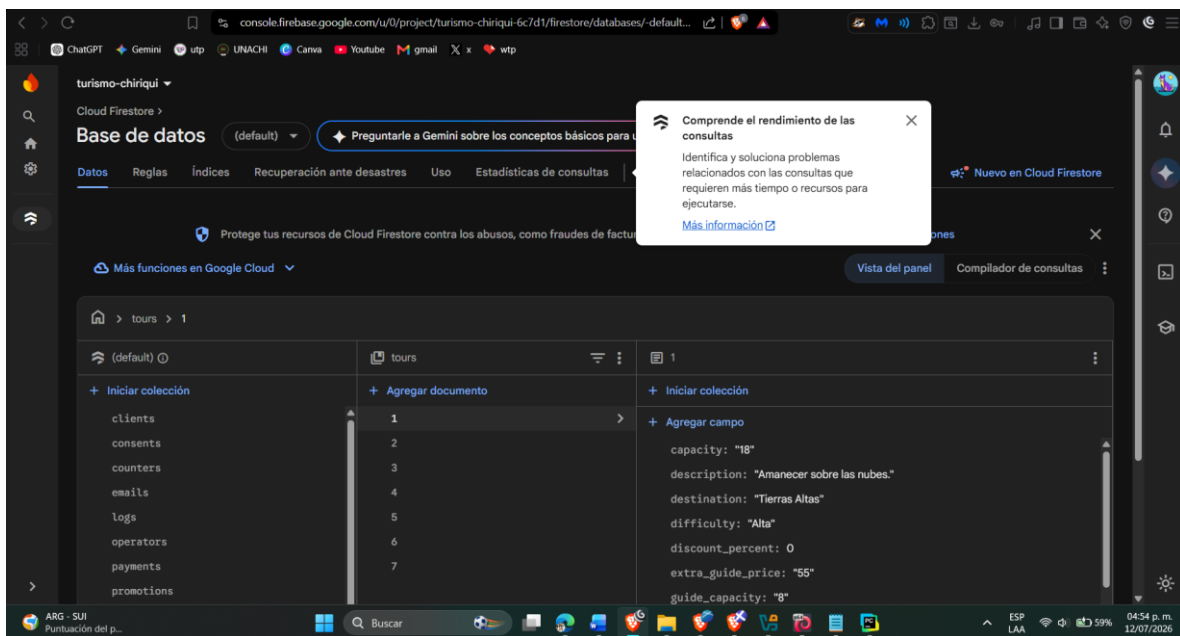
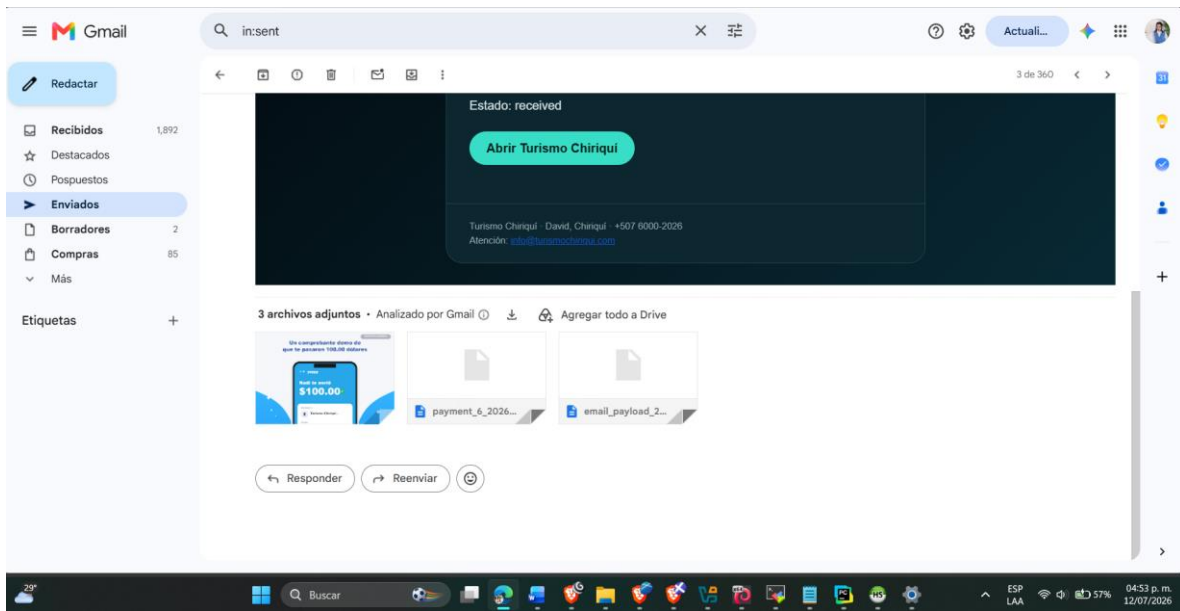
Empresas proveedoras

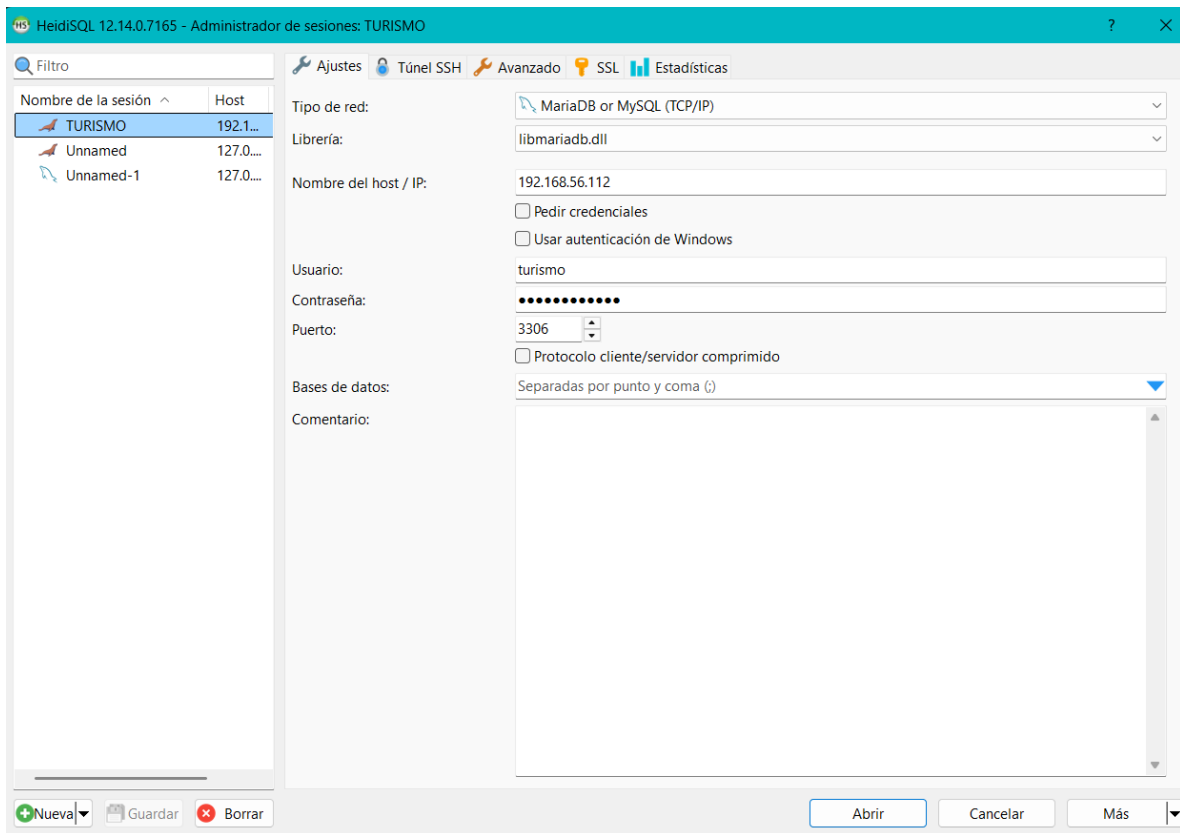
Nombre comercial	RUC o identificación fiscal	Representante legal
Correo	Teléfono	Dirección
Tipo de servicio	Descripción	Sitio web
activo		
Redes sociales	Mensaje de presentación	Estado
		activo
Logo o foto	Documento	
Seleccionar archivo Ningún archivo seleccionado	Seleccionar archivo Ningún archivo seleccionado	

Chiriquí Assistant

¡Hola! Pregúntame sobre tours, Yappy, horarios, reservas o qué llevar.

Escribe tu pregunta [Enviar](#)





II PARTE. Simulación de Red LAN. Valor 20 puntos

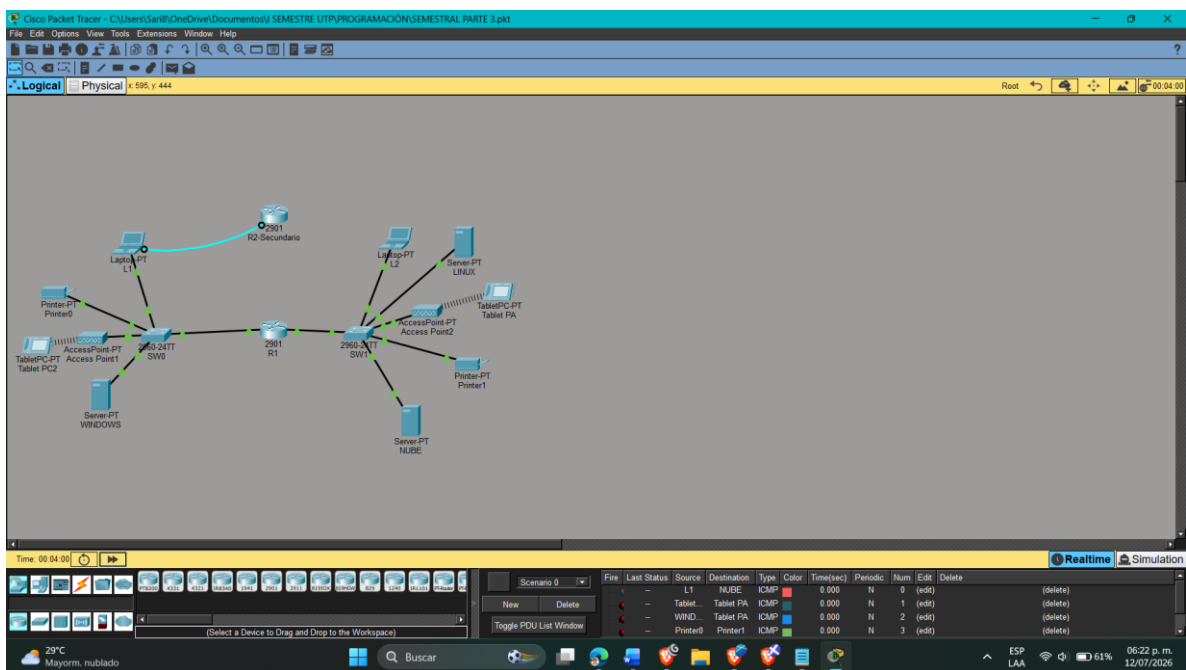


Tabla 1 Cuadro de requerimientos (IP)

Dispositivo	Interfaz	IP Address	Máscara de subred
R1	Gi0/0 (Suc. 1)	192.168.10.1	255.255.255.0
R1	Gi0/1 (Suc. 2)	192.168.20.1	255.255.255.0
R1 (HSRP virtual)	Gi0/0	192.168.10.254	255.255.255.0
R1 (HSRP virtual)	Gi0/1	192.168.20.254	255.255.255.0
R2	Gi0/0 (Suc. 1)	192.168.10.2	255.255.255.0
R2	Gi0/1 (Suc. 2)	192.168.20.2	255.255.255.0
SW0	VLAN 1	192.168.10.5	255.255.255.0
SW1	VLAN 1	192.168.20.5	255.255.255.0
Server WINDOWS	NIC	192.168.10.10	255.255.255.0
Server LINUX	NIC	192.168.20.10	255.255.255.0
Server NUBE	NIC	192.168.20.11	255.255.255.0
L1 (Laptop Suc. 1)	NIC	DHCP (192.168.10.20 – .200)	255.255.255.0
Printer0	NIC	DHCP (192.168.10.20 – .200)	255.255.255.0
Tablet PC2	WiFi	DHCP (192.168.10.20 – .200)	255.255.255.0
L2 (Laptop Suc. 2)	NIC	DHCP (192.168.20.20 – .200)	255.255.255.0
Printer1	NIC	DHCP (192.168.20.20 – .200)	255.255.255.0
Tablet PA	WiFi	DHCP (192.168.20.20 – .200)	255.255.255.0

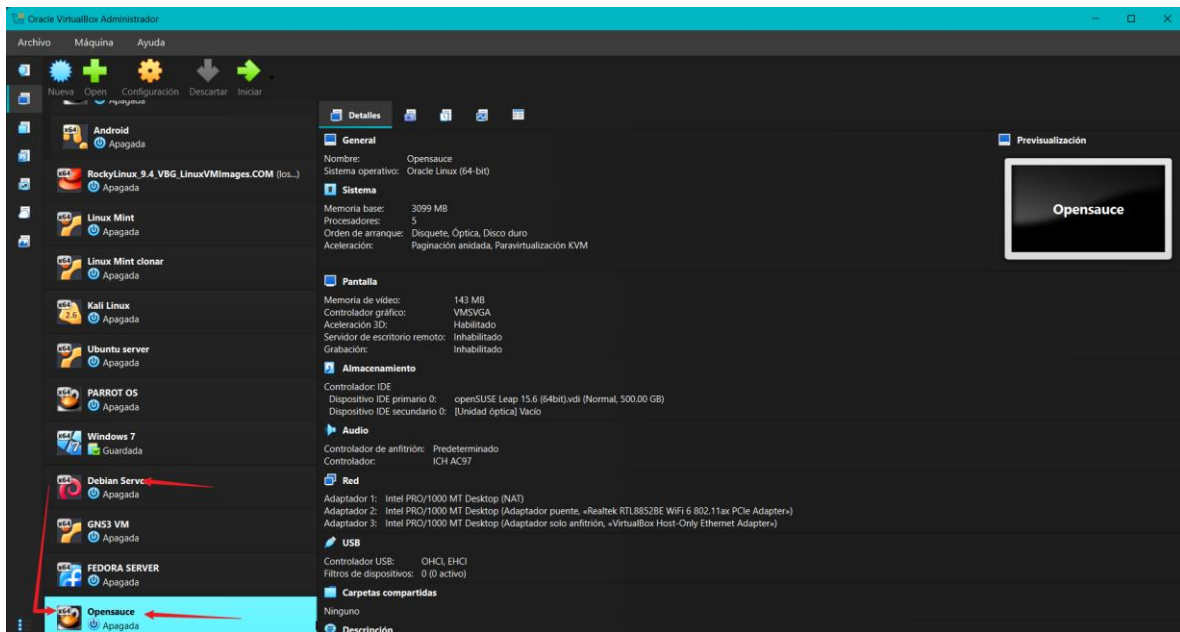
Tabla de credenciales

Servicio	Dispositivo(s)	Usuario	Contraseña
SSH / Enable	R1, R2, SW0, SW1	admin	Cisco12345
Enable secret	R1, R2, SW0, SW1	—	Cisco12345
Correo	gerencia@turismochiriqui.local	gerencia	12345
Correo	ventas@turismochiriqui.local	ventas	12345
Correo	reservas@turismochiriqui.local	reservas	12345
Dominio de correo	Server WINDOWS	—	turismochiriqui.local



SEMESTRAL PARTE
3.pkt

III PARTE. Caso de Estudio. Valor 15 puntos



```

192.168.56.112 (88)
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split Multitac Tunneling Packages Settings Help X server Exit

192.168.56.112 (88)
Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> SELECT User, Host, plugin
-> FROM mysql.user
-> WHERE User='turismo';
+-----+-----+-----+
| User | Host | plugin |
+-----+-----+-----+
| turismo | % | mysql_native_password |
+-----+-----+-----+
1 row in set (0.003 sec)

MariaDB [(none)]> CREATE USER 'turismo'@'localhost'
-> IDENTIFIED BY 'Turismo2026!';
Query OK, 0 rows affected (0.052 sec)

MariaDB [(none)]> GRANT ALL PRIVILEGES
-> ON turismo_db.*
-> TO 'turismo'@'localhost';
Query OK, 0 rows affected (0.028 sec)

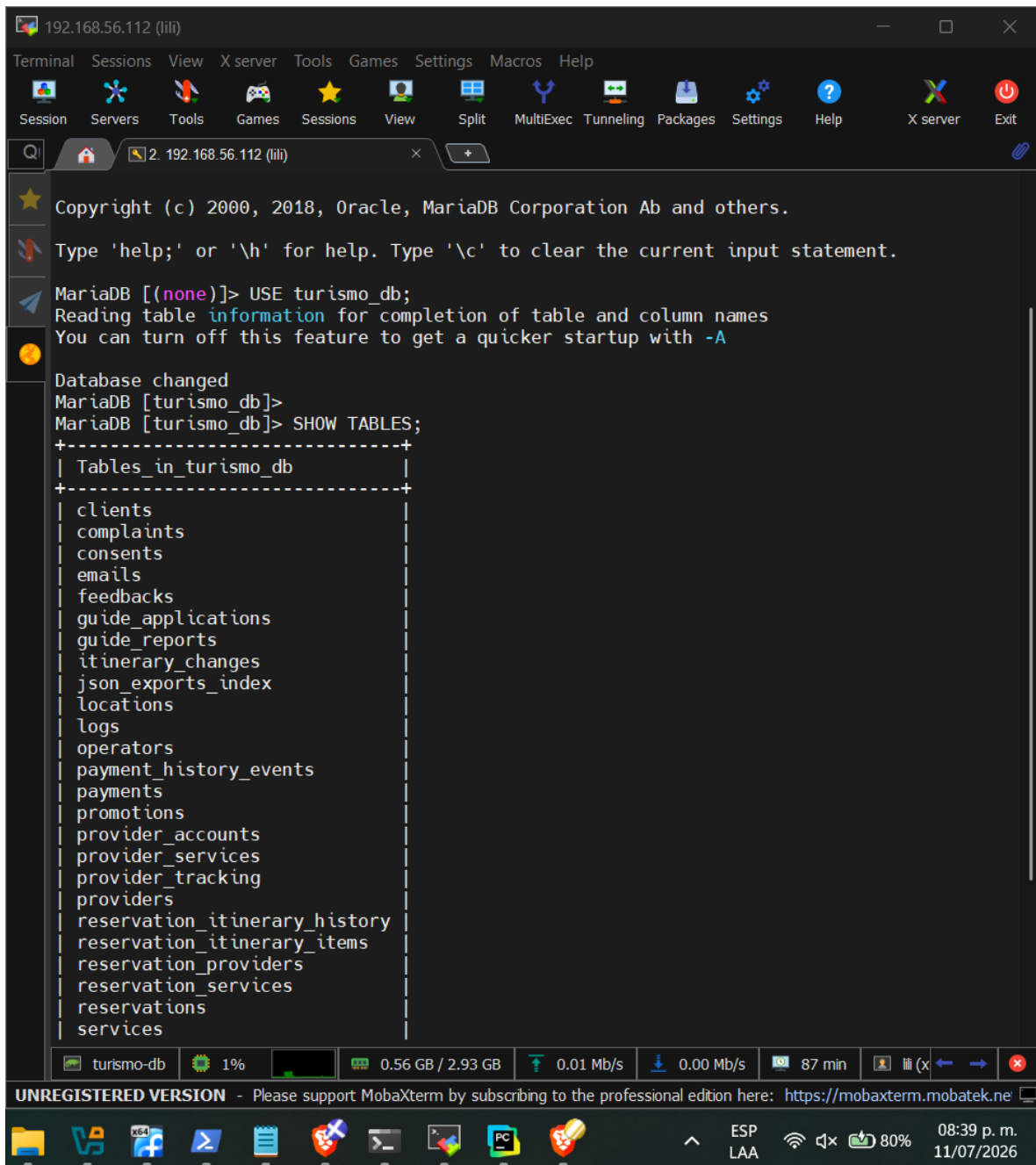
MariaDB [(none)]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]> exit;
Bye
lil@turismo-db:~$ mysql -u turismo -p
Enter password:
ERROR 1045 (28000): Access denied for user 'turismo'@'localhost' (using password: YES)
lil@turismo-db:~$ mysql -u turismo -p
Enter password:
Welcome to the MariaDB monitor. Commands end with ; or \g.
Your MariaDB connection id is 17
Server version: 10.11.15-MariaDB MariaDB package

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

```

UNREGISTERED VERSION - Please support MoboTerm by purchasing the professional edition here: <https://moboterm.mobotix.net/>



The screenshot shows a MobaXterm terminal window with the title bar '192.168.56.112 (lili)'. The menu bar includes Terminal, Sessions, View, X server, Tools, Games, Settings, Macros, and Help. The toolbar contains icons for Session, Servers, Tools, Games, Sessions, View, Split, MultiExec, Tunneling, Packages, Settings, Help, X server, and Exit. The terminal window has a tab labeled '2. 192.168.56.112 (lili)'. The terminal output is as follows:

```
Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> USE turismo_db;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MariaDB [turismo_db]>
MariaDB [turismo_db]> SHOW TABLES;
+-----+
| Tables_in_turismo_db |
+-----+
| clients               |
| complaints            |
| consents              |
| emails               |
| feedbacks             |
| guide_applications    |
| guide_reports         |
| itinerary_changes     |
| json_exports_index    |
| locations              |
| logs                  |
| operators             |
| payment_history_events |
| payments              |
| promotions            |
| provider_accounts     |
| provider_services     |
| provider_tracking     |
| providers             |
| reservation_itinerary_history |
| reservation_itinerary_items |
| reservation_providers |
| reservation_services  |
| reservations          |
| services              |
+-----+
```

At the bottom of the terminal window, there is a status bar showing 'turismo-db', '1%', '0.56 GB / 2.93 GB', '0.01 Mb/s', '0.00 Mb/s', '87 min', and a window icon. Below the terminal window, there is a banner for 'UNREGISTERED VERSION' with a link to the professional edition: <https://mobaxterm.mobatek.net>. The Windows taskbar at the bottom shows various icons and the system clock displaying '08:39 p.m. 11/07/2026'.

The screenshot shows a MobaXterm terminal window with the title bar '192.168.56.112 (lili)'. The terminal displays the output of a SQL query and the results of two database queries.

Query 1:

```

MariaDB [turismo_db]> select * from clients;
+-----+-----+-----+-----+-----+-----+-----+
| id | user_id | name           | email           | status | phone | welcome_coupon |
+-----+-----+-----+-----+-----+-----+-----+
| 1 |      2 | Andrea Martínez | cliente@email.com | active | NULL | NULL           |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.000 sec)

```

Query 2:

```

MariaDB [turismo_db]> select * from clients;
+-----+-----+-----+-----+-----+-----+-----+
| id | user_id | name           | email           | status | phone | welcome_coupon |
+-----+-----+-----+-----+-----+-----+-----+
| 1 |      2 | Andrea Martínez | cliente@email.com | active | NULL | NULL           |
| 3 |      9 | PRUEBAS        | sari821yt@hotmail.com | active | 7863602816 | NUEV010        |
+-----+-----+-----+-----+-----+-----+-----+
2 rows in set (0.001 sec)

```

The terminal window includes a menu bar (Terminal, Sessions, View, X server, Tools, Games, Settings, Macros, Help) and a toolbar with icons for Session, Servers, Tools, Games, Sessions, View, Split, MultiExec, Tunneling, Packages, Settings, Help, X server, and Exit. The status bar at the bottom shows 'turismo-db', '1%', '0.55 GB / 2.93 GB', '0.01 Mb/s', '0.00 Mb/s', '101 min', and system information: 'ESP LAA', '80%', '08:53 p. m.', '11/07/2026'.

```

192.168.56.107
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help X server Exit

(venv) turismo@turismo-db:~/turismo_chiriqui$ docker inspect mongo --format 'RestartPolicy: {{.HostConfig.RestartPolicy.Name}}'
RestartPolicy: unless-stopped
(venv) turismo@turismo-db:~/turismo_chiriqui$ docker ps --filter name=mongo
CONTAINER ID   IMAGE      COMMAND                  NAMES                  CREATED        STATUS        PORTS
ad0c1afa8540   mongo:4.4   "docker-entrypoint.s..."  mongo                 14 hours ago   Up 3 hours    0.0.0.0:27017->27017/tcp
(venv) turismo@turismo-db:~/turismo_chiriqui$ sudo iptables -L INPUT -n --line-numbers | head -10
Chain INPUT (policy ACCEPT)
num  target      prot opt source                destination            tcp dpts:40000:50000
1    ACCEPT      6    --  0.0.0.0/0             0.0.0.0/0             tcp dpt:21
2    ACCEPT      6    --  0.0.0.0/0             0.0.0.0/0             tcp dpt:22
3    ACCEPT      6    --  192.168.168.155       0.0.0.0/0             tcp dpt:22
4    ACCEPT      0    --  0.0.0.0/0             0.0.0.0/0
5    ACCEPT      6    --  192.168.56.0/24       0.0.0.0/0             tcp dpt:22
6    ACCEPT      6    --  192.168.56.0/24       0.0.0.0/0             tcp dpt:21
7    DROP        6    --  0.0.0.0/0             0.0.0.0/0             tcp dpt:22
8    DROP        6    --  0.0.0.0/0             0.0.0.0/0             tcp dpt:21
(venv) turismo@turismo-db:~/turismo_chiriqui$ sudo systemctl status knockd
● knockd.service - Port-Knock Daemon
   Loaded: loaded (/etc/systemd/system/knockd.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-07-12 14:36:26 EST; 2h 7min ago
     Docs: man:knockd(1)
    Main PID: 28509 (knockd)
      Tasks: 1 (limit: 5210)
     Memory: 464.0K
        CPU: 100ms
    CGroup: /system.slice/knockd.service
            └─28509 /usr/sbin/knockd -i enp0s8

Jul 12 14:36:26 turismo-db systemd[1]: Started knockd.service - Port-Knock Daemon.
Jul 12 14:36:26 turismo-db knockd[28509]: starting up, listening on enp0s8
Jul 12 14:36:46 turismo-db knockd[28509]: 192.168.168.155: openSSH: Stage 1
Jul 12 14:36:47 turismo-db knockd[28509]: 192.168.168.155: openSSH: Stage 2
Jul 12 14:36:47 turismo-db knockd[28509]: 192.168.168.155: openSSH: Stage 3
Jul 12 14:36:47 turismo-db knockd[28509]: 192.168.168.155: openSSH: OPEN SESAME
Jul 12 14:36:47 turismo-db knockd[28627]: openSSH: running command: /sbin/iptables -I INPUT
(venv) turismo@turismo-db:~/turismo_chiriqui$

Loading remote monitoring, please wait...
UNREGISTERED VERSION - Please support MobaXterm by subscribing to the professional edition here: https://mobaxterm.mobatek.net

```

192.168.56.112 (lili)

Terminal Sessions View X server Tools Games Settings Macros Help

Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help X server Exit

2. 192.168.56.112 (lili) 3. 192.168.56.107 4. 192.168.56.112 (lili)

```
turismo-db:~ # sudo systemctl status mariadb
● mariadb.service - MariaDB database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: disabled)
   Active: active (running) since Sun 2026-07-12 14:13:10 EDT; 3h 32min ago
     Docs: man:mysqld(8)
           https://mariadb.com/kb/en/library/systemd/
   Main PID: 1196 (mysqld)
    Status: "Taking your SQL requests now..."
     Tasks: 11 (limit: 3566)
      CPU: 27.307s
   CGroup: /system.slice/mariadb.service
           └─1196 /usr/sbin/mysqld --defaults-file=/etc/my.cnf --user=mysql --socket=/

Jul 12 14:12:59 turismo-db systemd[1]: Starting MariaDB database server...
Jul 12 14:13:01 turismo-db mysql-systemd-helper[1196]: 2026-07-12 14:13:01 0 [Warning] f
Jul 12 14:13:10 turismo-db systemd[1]: Started MariaDB database server.
turismo-db:~ # sudo firewall-cmd --get-active-zones
sudo firewall-cmd --zone=dmz --list-all
sudo firewall-cmd --zone=public --list-all
dmz
  interfaces: eth3
docker
  interfaces: docker0
public (default)
  interfaces: eth1 eth2
turismo-db:~ # sudo systemctl status knockd
● knockd.service - Port-Knocking Daemon
   Loaded: loaded (/usr/lib/systemd/system/knockd.service; enabled; preset: disabled)
   Drop-In: /etc/systemd/system/knockd.service.d
           └─override.conf
   Active: active (running) since Sun 2026-07-12 17:24:37 EDT; 21min ago
     Main PID: 20125 (knockd)
       Tasks: 1 (limit: 3566)
        CPU: 3.097s
     CGroup: /system.slice/knockd.service
           └─20125 /usr/sbin/knockd -i eth3

Jul 12 17:36:01 turismo-db knockd[25488]: success
Jul 12 17:36:52 turismo-db knockd[20125]: 192.168.168.155: opencloseSSH: Stage 1
Jul 12 17:36:53 turismo-db knockd[20125]: 192.168.168.155: opencloseSSH: Stage 2
```

turismo-db 1% 0.59 GB / 2.93 GB 0.01 Mb/s 0.00 Mb/s 213 min

UNREGISTERED VERSION - Please support MobaXterm by subscribing to the professional edition here: <https://mobaxterm.mobatek.net>

ESP LAA 49% 04:46 p. m. 12/07/2026

